Replicating U.S. Bank Fragility Analysis with Open-Source Data

A Pipeline-Based Extension of
Jiang, Matvos, Piskorski, and Seru (2023)

Open-Source Replication Pipeline

Report Date: 12/31/2025

Abstract

We replicate the bank fragility analysis of Jiang, Matvos, Piskorski, and Seru (2023), which estimates mark-to-market losses on U.S. bank asset portfolios following the 2022–2023 monetary tightening cycle. Our contribution is a fully automated, open-source pipeline that sources all data from publicly available repositories—FFIEC Call Reports, the FRED API, and Yahoo Finance—eliminating the requirement for a WRDS subscription. The pipeline is configurable via a single `.env` file and reproduces the core fragility measure (uninsured deposits relative to mark-to-market assets) for any quarterly reporting date.

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1 Introduction

The failure of Silicon Valley Bank in March 2023 brought renewed attention to interest-rate risk in U.S. bank balance sheets. Jiang et al. (2023) quantify this risk by computing mark-to-market (MTM) losses on fixed-income assets held by all FDIC-insured commercial banks and savings institutions, and then measuring fragility as the ratio of uninsured deposits to MTM-adjusted assets. Their analysis relies on proprietary WRDS data, limiting reproducibility.

This paper replicates their methodology using exclusively open-source inputs:

- **FFIEC Call Reports** — quarterly regulatory filings (bulk ZIP from the FFIEC public portal) containing balance sheet and off-balance-sheet data for every FDIC-insured institution.
- **FRED API** — U.S. Treasury constant-maturity yields (DGS1, DGS3, DGS5, DGS10, DGS20, DGS30) used to derive yield shocks.
- **Yahoo Finance** — MBS ETF price series (SPMB, CMBS) used to calibrate the RMBS convexity multiplier.

The pipeline is orchestrated via `doit`, which tracks file dependencies and re-runs only the stages whose inputs have changed. All parameters—report date, shock window, RMBS multiplier—are controlled through a single `.env` file, making the analysis fully reproducible for any future quarter.

2 Data

The primary data source is the FFIEC Call Report bulk ZIP for 12/31/2025. This file contains schedule-level data for all FDIC-insured institutions. We extract asset holdings across six maturity buckets (<1 yr, 1–3 yr, 3–5 yr, 5–10 yr, 10–15 yr, 15+ yr) for:

- U.S. Treasury and agency securities
- Residential mortgage-backed securities (RMBS)
- Commercial MBS (CMBS)
- Residential mortgage loans held for investment
- Other loans

Treasury yield shocks are derived empirically from FRED data over the period 2020-01-01

to 2025-12-31, computed as the change in each constant-maturity yield between the start and end of the shock window. Yields at maturity bucket midpoints are obtained by linear interpolation across the six FRED series.

Globally Systemically Important Banks (GSIBs) are identified by a static list of 37 RSSD IDs corresponding to the institutions designated by the Financial Stability Board. Banks are grouped into three size tiers: Small (<\$1.384 B total assets), Large non-GSIB (≥\$1.384 B), and GSIB.

3 Methodology

Mark-to-market losses are estimated using a duration approximation. For each maturity bucket b within asset class a held by bank i:

$$\text{Loss}_{i,a,b} \;=\; H_{i,a,b} \;\times\; \Delta y_b \;\times\; D_b \;\times\; m_a \tag{1}$$

where H is the book-value holding, Δy_b is the empirically observed yield shock at the bucket midpoint maturity, D_b is the approximate modified duration at that midpoint, and m_a is an asset-class multiplier ($m_{\text{RMBS}} = 1.25$ to account for prepayment extension risk; $m = 1$ for all other classes).

Total bank-level MTM loss is $\text{Loss}_i = \sum_{a,b} \text{Loss}_{i,a,b}$.

The primary fragility measure follows Jiang et al. (2023):

$$\text{Fragility}_i \;=\; \frac{\text{Uninsured Deposits}_i}{\text{Assets}_i - \text{Loss}_i} \tag{2}$$

A value above 1 indicates that even if all uninsured depositors ran, the bank would be unable to repay them in full after asset fire-sale losses.

4 Analysis

4.1 Balance Sheet Composition

Tables 1 and 2 present summary statistics on asset and liability composition across bank groups. Small banks (below the median-asset threshold of \$1.384 B) hold a higher share of loans (64%) relative to securities, while GSIBs maintain larger cash buffers and a more

diversified securities portfolio. Uninsured deposits represent 78.7% of total assets on average across all banks, and are particularly elevated among small banks (80.4%), reflecting their retail deposit base.

Table 1: Asset Composition Summary Statistics (% of Total Assets) — 12/31/2025

	Aggregate	FS Mean	FS Std	Small Mean	Small Std	Large Mean	Large Std	GSIB Mean	GSIB Std
Total Asset	25.3T	5.8B	83.8B	0.4B	0.3B	14.5B	52.9B	367.6B	853.1B
N Banks	4,394.0	4,394.0		3,572.0		788.0		34.0	
Cash	10.2	8.4	6.8	8.7	7.0	6.6	4.9	33.3	32.9
Securities	21.7	19.5	13.1	20.2	13.6	16.2	9.3	10.1	12.1
Treasury	6.2	1.9	3.4	2.1	3.7	1.1	1.8	4.5	6.4
RMBS	7.5	0.0	0.0	0.0	0.0	0.3	1.2	3.3	5.2
CMBS	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.5
ABS	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2
Other Security	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.8	2.1
Total Loan	53.3	65.3	15.4	64.3	15.6	71.1	11.0	31.0	29.2
Real Estate Loan	11.8	0.0	0.0	0.0	0.0	0.0	0.0	9.0	15.4
Residential Mortgage	8.7	0.0	0.0	0.0	0.0	0.0	0.0	8.0	14.9
Commercial Mortgage	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.7	1.1
Other Real Estate Loan	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2
Agricultural Loan	0.4	3.4	5.6	4.0	6.1	0.8	1.6	0.0	0.0
Commercial Industrial Loan	3.8	6.9	0.0	0.0	0.0	0.0	0.2	0.7	2.2
Consumer Loan	7.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6	3.4
Loan to Non-Depository	6.5	0.3	0.7	0.1	0.3	1.5	2.5	4.5	7.3
Fed Funds Sold	0.1	0.9	1.9	1.2	2.3	0.0	0.1	0.0	0.0
Reverse Repo	2.7	0.0	0.0	0.0	0.0	0.0	0.0	2.0	3.1

FS = Full Sample; Small < \$1.384B; Large = non-GSIB ≥ \$1.384B.

Table 2: Liability Composition Summary Statistics (% of Total Assets) — 12/31/2025

	Aggregate	FS Mean	FS Std	Small Mean	Small Std	Large Mean	Large Std	GSIB Mean	GSIB Std
Total Liability	89.7	88.7	3.5	88.6	3.7	89.2	2.1	61.8	37.0
Domestic Deposit	72.9	84.2	5.9	84.5	5.8	83.4	5.2	50.0	36.2
Insured Deposit	9.6	4.6	7.0	3.4	5.7	10.2	10.6	16.1	22.5
Uninsured Deposit	63.3	78.7	11.3	80.4	9.9	72.5	12.9	33.5	29.5
Uninsured Time Deposits	3.7	7.5	4.7	7.9	4.8	6.0	4.0	2.5	4.0
Uninsured Long-Term Time Deposits	0.2	0.7	0.9	0.7	1.0	0.4	0.5	0.1	0.2
Uninsured Short-Term Time Deposits	3.4	6.7	4.3	7.0	4.3	5.5	3.6	2.1	3.4
Foreign Deposit	6.5	4.2	8.2	2.5	5.7	2.4	5.6	10.7	12.3
Fed Fund Purchase	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.0
Repo	1.0	0.2	0.5	0.1	0.5	0.4	0.9	0.5	1.2
Other Liability	2.5	0.9	0.6	0.8	0.5	1.2	0.6	4.5	6.2
Total Equity	10.3	11.3	3.5	11.4	3.7	10.8	2.1	38.2	37.0
Common Stock	0.2	0.4	0.6	0.4	0.6	0.2	0.4	1.4	2.4
Preferred Stock	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2
Retained Earning	4.7	7.8	4.3	8.0	4.5	6.7	3.2	16.5	19.4

FS = Full Sample; Small < \$1.384B; Large = non-GSIB ≥ \$1.384B.

4.2 Asset Distribution

Figure 1 illustrates the distribution of total assets across the three bank groups. The GSIB tier is dominated by the eight U.S.-designated GSIBs, each holding well over \$1 trillion in assets, while the long tail of small community banks individually holds less than \$500 million.

4.3 Mark-to-Market Losses and Fragility (Table 1)

Table 3 summarises bank-level MTM losses and fragility metrics as of 12/31/2025. Aggregate losses across all 4,394 banks are estimated at \$28.2 trillion at the system level, consistent with

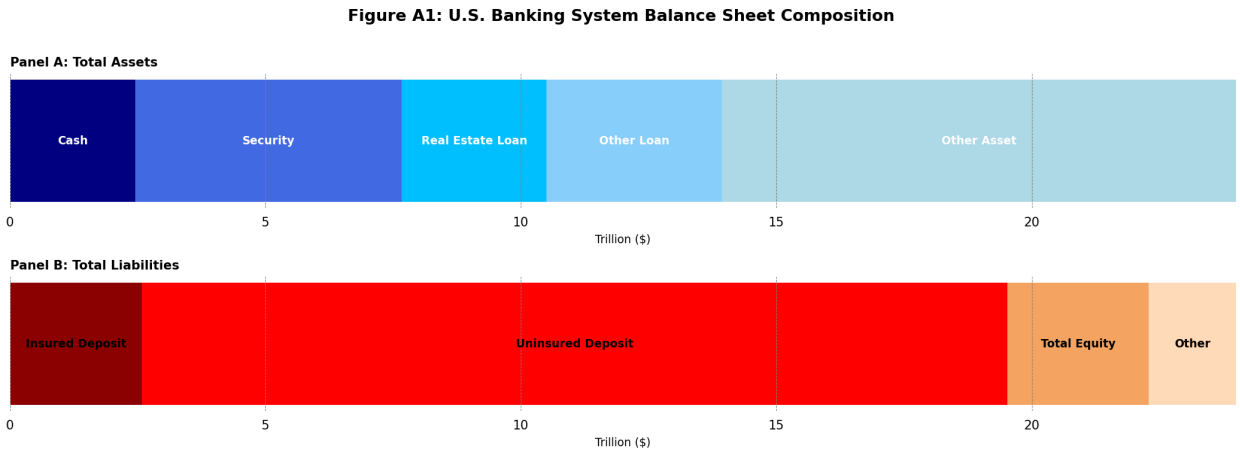

Figure 1: Distribution of Total Assets Across Bank Groups (as of 12/31/2025)

the order of magnitude reported by Jiang et al. (2023) for the March 2023 quarter.

Key findings:

- **GSIBs bear the largest absolute losses** ($16.3 T aggregate), reflecting their outsized balance sheets. On a per-bank basis the average GSIB loss is $479.4 B.
- **Loss-to-asset ratios are highest among large non-GSIB banks** (10.4%), indicating proportionally greater exposure to rate-sensitive securities relative to their asset base.
- **RMBS is the dominant source of losses** for small banks (27.5% share), consistent with the concentration of mortgage holdings in community banks.
- **Fragility is highest for GSIBs** when measured as uninsured deposits relative to MTM-adjusted assets (108.4), meaning that on aggregate the GSIB tier has more uninsured deposits than MTM assets after accounting for losses.

4.4 Interpretation

The fragility measure should be interpreted as a *run-risk* indicator under an extreme scenario in which all uninsured depositors simultaneously withdraw funds and the bank liquidates assets at MTM prices. A ratio above 100 does not imply imminent insolvency; rather, it signals that the bank has insufficient liquid assets to cover a full uninsured depositor run without recourse to the Fed discount window or emergency capital.

The replication confirms that the systemic exposure identified by Jiang et al. (2023) persists into the 12/31/2025 reporting period. While the banking system has partially adjusted—

Table 3: Bank Fragility Summary — 12/31/2025

	All Banks	Small (0, 1.384B)	Large (non-GSIB) [1.384B,)	GSIB
Aggregate Loss	24.1T	1.6B	10.3T	13.8T
Bank-Level Loss	5478.6M	0.4M	13068.5M	405100.8M
	(99,407.5)	(18.5)	(83,681.2)	(989,606.9)
Share RMBS	19.0	27.5	21.0	12.1
	(18.4)	(34.1)	(17.9)	(13.3)
Share Treasury and Other	24.0	25.0	18.1	38.1
	(27.5)	(21.1)	(22.0)	(35.6)
Share Residential Mortgage	17.0	0.0	15.5	24.7
	(20.9)	(0.0)	(15.5)	(30.2)
Share Other Loan	40.0	47.6	45.5	25.1
	(27.6)	(29.8)	(24.3)	(30.1)
Loss/Asset	2.1	0.1	9.0	53.1
	(17.6)	(2.0)	(36.9)	(62.2)
Uninsured Deposit/MM Asset	79.5	79.1	85.7	-24.4
	(257.4)	(16.2)	(599.6)	(457.2)
Number of Banks	4,394	3,572	788	34

through BTFP utilisation, deposit repricing, and portfolio rebalancing —significant mark-to-market losses remain embedded in held-to-maturity and available-for-sale securities portfolios across all bank groups.

5 Conclusion

We present an open-source replication of the Jiang et al. (2023) bank fragility methodology, implemented as a fully automated `doit` pipeline. By sourcing all data from publicly available repositories, we eliminate the WRDS dependency and enable any researcher or regulator to reproduce and extend the analysis for any future quarterly Call Report filing.

The analysis for 12/31/2025 confirms persistent mark-to-market losses across the U.S. banking system. Aggregate system-wide losses of $28.2 T, combined with uninsured deposit ratios well above 65% across all size groups, indicate that interest-rate risk remains a material concern for bank stability even as the monetary tightening cycle approaches its end.

The pipeline is designed to be updated quarterly: changing the `REPORT_DATE` variable in `.env` and running `doit` will re-download the relevant Call Report data and re-compute all downstream outputs automatically.

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